

Final HRI Dataset Table

#	Split	Context	Scenario	Emotion	Gesture	Motion	Missing	Intent label	Sub Intent label	Research goal	Why rule-based fails / cue justification	Robot Action	Test Case
1	Train	Classroom	Student neutral, raises hand, sitting	neutral	raise hand	sitting	—	F04 Help request	S01 Ask a question	G1	All 4 cues needed. Raise hand + sitting + classroom + neutral emotion confirms need help.	Answer the question	-
2	Train	Classroom	Student waves at robot while walking toward	happy	wave	Walk (toward)	—	F01 Greet & acknowledge	S02 Greet	G1, G3	Wave + happy could mean greeting. Motion direction is the critical disambiguator. Rule: wave→greet fires incorrectly here without motion context.	Acknowledge greeting and prepare for interaction	-
3	Train	Classroom	Student calmly writes notes with neutral face during lecture	neutral	Point / (writing)	sitting	—	F05 Neutral pass	S03 Focus on work	G1, G3	Rule: neutral+sitting→idle, safe to interrupt. Wrong. Gesture (writing/pointing at notes) + classroom = focused work. All 4 cues needed to suppress robot action.	Maintain awareness	-
4	Train	Classroom	Student sad expression, barely moving	sad	thumbs down	sitting	—	F04 Help request	S04 Don't Understand	G1	Emotion+gesture+motion all align. Context (classroom not home) distinguishes academic distress from personal grief.	Respond with calm supportive behavior	-
5	Train	Classroom	Student raises both hands in frustration, rigid posture.	angry	both hands up	stand	—	F02 Alert	S05 Indicate Technical Problem	G1, G3	Rule: angry→intervene immediately. But stand+classroom context = controlled	Pause task and provide troubleshooting assistance	-

											frustration, not physical threat. Transformer learns deferred calm response.		
6	Train	Classroom	Student pushes worksheet away with disgust, turns body away from desk	disgust	thumbs down	stepping back	—	F08 Ask for a break/relief	S06 Ask for a break	G1, G3	Rule: disgust→check for smell/food. Completely wrong in classroom. Context flips meaning to task aversion. Robot offers alternative activity.	Acknowledge request and suggest appropriate break procedure	-
7	Train	Classroom	Student beckons robot with neutral expression while sitting	neutral	beckoning	sitting	—	F03 Task assist	S07 Seek individual help	G1, G3	Rule: neutral→no action. Misses beckoning gesture entirely. All 4 cues required — neutral emotion would suppress response in rule system.	Approach student and provide localized assistance	-
8	Train	Classroom	Student carrying books	Neutral	[MISSING]	walking	Gesture	F06 Give way	S08 Need the path to be cleared	G2		Move aside promptly and reopen safe navigation space	
9	Train	Classroom	Student suddenly, stand up from chair	surprise	both hands up	stand	—	F02 Alert	S09 Emergency	G1	All 4 cues fire together. Both-hands-up + stand + surprise + classroom = alarm state. Robot halts and checks surroundings.	Halt risky action and ensure safety	
11	Train	Classroom	Student raises hand smiling while sitting	happy	Raise hand	sitting	—	F05 Neutral pass	S09 Volunteer an Answer	G3		Give a chance	
12	Train	Classroom	Student gives thumbs up	happy	thumbs up	sitting	—	F01 Greet & acknowledge	S10 Signal/Confirm Understanding	G1, G3	Rule: thumbs up→success always. But thumbs up + disgust = sarcasm (see test set). Here emotion+gesture align — transformer learns emotion resolves gesture ambiguity.	Continue	

13	Test	Classroom	Student raises hand with neutral expression - emotion sensor occlusion	[MISSING]	raise hand	leaning forward	emotion	F04 Help request	S11 Not understanding the lesson	G2	Emotion missing. Gesture + motion + classroom context still sufficient. Tests if model redistributes attention to remaining 3 cues correctly.	Ask questions to clarify	T01/T04, T03, T05
14	Test	Classroom	Student slumped, sad, staring - no clear hand gesture captured	sad	[MISSING]	sitting	gesture	F04 Help request	S12 Signal Confusion	G2	Gesture missing (hands below camera). Emotion + posture motion + classroom still maps to help request. Tests graceful degradation.		T01/T04, T03, T05
15	Test	Classroom	Student gives thumbs up with disgusted expression	disgust	thumbs up	sitting	—	F08 Ask for a break/relief	S06 Ask for a break	G3	Critical rule-based failure. Rule: thumbs up→success→acknowledge. Transformer: disgust overrides thumbs up = sarcastic completion, offer relief. Requires emotion+gesture interaction.		T01/T04, T02, T05
16	Test	Classroom	Student stands up angrily and points at robot, stepping toward it	angry	point	walk	—	F06 Give way	S12 Reject Assistance	G1, G3	Rule: angry+point→alert/intervene. But motion direction (walking TOWARD robot) + point at robot = back off intent. Context + motion direction critical.		T01/T04, T02, T05
17	Test	Classroom	Student waves with neutral face while walking out - no emotion clearly detected	[MISSING]	wave	walk	emotion	F09 Acknowledge farewell	S13 Farewell	G2, G3	Emotion missing. Wave + walk direction + classroom-exit context sufficient. Rule would fire greet instead. Tests motion-direction		T01/T04, T03, T05

											reasoning without emotion.		
18	Train	Kitchen	Cook smiles, gives thumbs up, stands still after tasting completed dish	happy	thumbs up	stand	—	F01 Greet & acknowledge	S01 Task Success	G1	All 4 cues confirm task success. Context (kitchen) + stationary motion + thumbs up + happy = genuine completion signal. No ambiguity.	A01 Acknowledge & prompt next task	-
19	Train	Kitchen	Person extends both hands out as stop gesture and runs backward from smoke	fear	both hands up	Move backward (run)	—	F02 Emergency — halt & alert	S02 Environmental Hazard S03 Physical Injury	G1	Highest-priority scenario. All 4 cues maximally active. Fear + run + both hands = unambiguous emergency. Rule partially works but fails when any cue is noisy.	A02 Halt, fire/smoke alert A03 Halt, medical alert	-
20	Train	Kitchen	Person beckons robot toward stove area with neutral expression while walking there	neutral	beckoning	walk	—	F03 Task assist	S04 Cooking Assist	G1, G3	Rule: neutral→no action. Beckoning gesture ignored. Transformer: neutral + beckoning + walk-toward-object + kitchen = assistance requested at stove.	A04 Follow to stove, await instruction	-
21	Train	Kitchen	Person slumped against counter, sad, barely moving, hands down	sad	thumbs down	sitting	—	F04 Help request	S05 Task Help	G1	Context (kitchen) distinguishes cooking exhaustion from general distress. Emotion+motion+gesture align on distress. Robot approaches.	A05 Offer task guidance	-
22	Train	Kitchen	Person stands calmly chopping with neutral expression, back to robot	neutral	point	stand	—	F05 Neutral pass	S06 Focused Task	G1, G3	Rule: neutral+stand→available. Wrong. Focused task motion (chopping) + kitchen = do not interrupt. All 4 cues needed to suppress robot intervention.	A06 Hold position, do not interrupt	-

23	Train	Kitchen	Person recoils in disgust, steps back from refrigerator	disgust	thumbs down	stepping back	—	F08 Offer break or relief	S07 Ask for a break	G1, G3	Rule: disgust+kitchen→check food/smell. Partially correct but maps to wrong robot action. Transformer: disgust + stepping back = person needs relief, not robot inspection.	A07 Acknowledge request and suggest appropriate break procedure	-
24	Train	Kitchen	Person angrily slams hand on counter, rigid posture, standing	angry	both hands up	stand	—	F07 Alert & de-escalate	S08 Task Frustration	G1, G3	Rule: angry→intervene immediately. Kitchen context + rigid stand (not fleeing) = frustration, not physical danger. Transformer applies calm de-escalation, not emergency halt.	A08 Offer calm assistance, soft tone	-
25	Train	Kitchen	Person waves at robot while walking toward exit	happy	wave	walk	—	F09 Acknowledge farewell	S09 Session-End Farewell S10 Temporary Leave	G1, G3	Wave + happy could be greeting. Walk direction (toward exit) + session-end context = farewell. Rule: wave→greet fires wrong action.	A09 Wave back, do not follow A10 Acknowledge, hold position	-
26	Train	Kitchen	Person raises both hands startled when hot pan is touched	surprise	both hands up	stepping back	—	F02 Emergency — halt & alert	S02,S03	G1	Surprise + both hands + stepping back + kitchen = burn/burn-risk emergency. Context critical — same cues in classroom = different intent.	A02,A03	-
27	Train	Kitchen	Person pushes robot aside with disgusted expression, turning away	disgust	point	stepping back	—	F06 Give way	S11 Obstacle Clearance	G1, G3	Rule: disgust+stepping back→hazard check. But gesture directed at robot + stepping back = give way. Same as classroom case but kitchen context — shows	A11 Move aside, clear path	-

											cross-environment generalization.		
28	Train	Kitchen	Person sits down looking sad, slow movement, staring at failed dish	sad	thumbs down	sitting	—	F10 Encourage & motivate	S12 Cooking Failure	G1	Sad + sitting + no reaching gesture = giving up, not requesting help. Distinguishes F04 from F10 by absence of directed gesture. Kitchen context confirms cooking failure context.	A12 Encourage, suggest alternative recipe	-
29	Train	Kitchen	Person points at shelf while walking toward it with neutral expression	neutral	point	walk	—	F03 Task assist	S13 Carry Assistance	G1, G3	Rule: neutral→no action. Point + walk-toward + kitchen = directing robot to fetch item from shelf. All 4 cues required — emotion suppression in rule system causes miss.	A13 Follow, offer to carry	-
30	Test	Kitchen	Person runs away from area with fear — gesture not captured (fast motion blur)	fear	[MISSING]	run	gesture	F02 Emergency — halt & alert	S02,S03	G2	Gesture lost due to motion blur. Fear + run + kitchen = emergency still detectable from 3 cues. Tests safety-critical robustness — model must not fail here.	A02,A03	T03, T05
31	Test	Kitchen	Person gives thumbs up with angry expression after being told to redo a recipe	angry	thumbs up	stand	—	F07 Alert & de-escalate	S08	G3	Rule: thumbs up→success→acknowledge. Transformer: angry + thumbs up = sarcastic compliance, rising tension. Emotion overrides gesture. Strongest rule-based failure case.	A08	T02,T05

32	Test	Kitchen	Person beckons robot with sad expression — context sensor offline	sad	beckoning	stand	context	F04 Help request	S05	G2	Context missing. Sad + beckoning + stand could be kitchen or office. Model must use emotion + gesture to infer help request even without environment confirmation.	A05	T03
33	Test	Kitchen	Person raises hand and smiles while sitting at kitchen table after eating	happy	raise hand	sitting	—	F01 Greet & acknowledge	S01	G3	Rule: raise hand→student participation (classroom rule). In kitchen context, raise hand + happy + sitting = social acknowledgment/greeting. Context flips the meaning entirely.	A01	T04,T05
34	Test	Kitchen	Person stands neutral, no gesture visible, stationary — emotion sensor noisy	[MISSING]	[MISSING]	stand	emotion, gesture	F05 Neutral pass	S06	G2	Two cues missing. Only motion + context remain. Tests worst-case degradation. Model should default to neutral pass (safe fallback) rather than misfire an active intent.	A06	T03
35	Train	Kitchen	Person wipes hands on apron, neutral face, looks around kitchen — task transition moment	Neutral	Point (self-directed wiping motion)	Stand	-	F05 Neutral pass	S14 Task End	G1,G3	Rule: neutral+stand → available for interaction → approach. Misinterprets transition as invitation. Gesture (inward self-directed motion, not outward toward robot or object) + neutral emotion = do not engage	A06	T05
36	Train	Kitchen	Person angrily points finger repeatedly at spilled food on	Angry	Point (directed at floor/mess)	Stand	-	F07 Alert & de-escalate	S08	G1,G3	Rule: angry+point → threat response → back away. Completely wrong.	A08	

			floor, looking at robot								The point is directed at the FLOOR (task object), not at the robot's personal space. Motion (stationary) + kitchen context = complaint, not aggression.		
37	Train	Kitchen	Person slowly stirs pot with content neutral expression, humming - deeply focused on cooking	Neutral	Point (stirring motion)	Stand	-	F05 Neutral pass	S06	G1,G3	Rule: neutral+stand → approach and offer help. Interrupting a focused cooking task is unwanted and potentially dangerous near hot surfaces. All 4 cues needed to confirm 'do not disturb' state.	A06	
38	Test	Kitchen	Person stands frozen with wide eyes staring at oven — no gesture, emotion sensor fails	[MISSING]	Point (frozen stare direction)	Stand (frozen)	Emotion	F02 Emergency — halt & alert	S02,S03	G2	Rule: requires fear/surprise emotion to trigger emergency. Without emotion cue, rule defaults to F05 neutral pass — dangerously wrong near oven. Fusion uses frozen motion + stare-gesture + kitchen-oven context to escalate.	A02,A03	T03,T05
39	Test	Kitchen	Person gives thumbs down while smiling and shaking head playfully — tasting game with family	Happy	Thumbs down	Stand	-	F01 Greet & acknowledge	S08	G3	Rule: thumbs-down → distress/failure → F04 help request or F08 offer relief. Completely wrong. Happy emotion is the critical override. Transformer must learn emotion-gesture interaction: happy+thumbs-down = playful,	A08	T02,T05

											sad+thumbs-down = genuine negative.		
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